

UNIVERSITY OF ILLINOIS AT CHICAGO

Department of Chemical Engineering

CHE 396: Senior Design I

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Homework 4

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Project Charter

Title	Goal	In-Scope	Out-of-Scope
<i>Economic Break-down for the production of MEK</i>	Calculate costs and revenues associated with the production of MEK.	Forecasting of the net present value of the plant.	- Optimization of the process - Location of the manufacturing facilities
Deliverables		Team Members	
- Fixed capital cost - Cash cost of production - Price forecasting for raw materials - Sensitivity analysis on utilities, raw materials and product cost - Net present value of the MEK plan		- Yier Li - Anne Serban - Maurice Smith - Melissa Djonlich	

Design Plan

Task	Start Date	Duration (Days)	Finish Date	Responsible Party
Task Selection	11/12	0.5	11/12	All
Design				
Layout	11/12	3.0	11/15	All
Design Plan	11/12	3.0	11/15	All
Implementation				
Introduction	11/12	1.0	11/13	All
Capital Cost with Order of Magnitude	11/12	5.0	11/17	All
Hand Calculations	11/12	3.0	11/15	All
Aspen Calculations	11/12	3.0	11/15	All
Inside Battery Limits (ISBL)	11/12	3.0	11/15	All
Offsite Battery Limits (OSBL)	11/12	3.0	11/15	All
Contingency and Engineering Costs	11/12	3.0	11/15	All
Variable Cost of Production (VCOP)	11/12	3.0	11/15	All
Fixed Cost of Production (FCOP)	11/12	3.0	11/15	All
Revenues and Gross Profit	11/12	3.0	11/15	All
Simple Pay-Back Period and Return on Investment (ROI)	11/12	3.0	11/15	All
Net Present Value (NPV)	11/12	4.0	11/16	All
Discounted Cash Flow Rate of Return (DCFROR)	11/12	4.0	11/16	All
Sensitivity Analysis	11/12	5.0	11/17	All

Introduction

When performing the economic analysis for a process, several factors are investigated and calculated. These factors include capital costs, material costs, utilities, variable and fixed operating costs. Capital costs, which encompass the inside battery limits and off-site battery limits. These costs are the total costs for purchasing unit operations pertinent to the production of the process, the hiring of engineers to design and landscape the process and contingency. Material costs are the cost associated with the purchase of any and all raw materials minus any credits for side-products that are produced and could be sold for profit. Utilities are the costs associated with running the plant on a daily basis, this includes things like electricity, gas, steam, etc. The aforementioned costs are associated with the variable cost of production, or VCOP. These prices fluctuate with the increase or decrease with the amount of product a plant is expected to produce. Costs associated with the operation of the plant that are not considered utilities, such as employees, insurance, maintenance and non-revenue-based taxes are classified as fixed cost of production. All of these costs were calculated and used in the determination of the gross profit, net profit, and net present value (NPV). Prices of MEK, trichloroethane, 2-butanol were all taken from “Gavin’s Book”. These prices were assumed valid for 2019, subsequently all calculations were based on these figures. Variations in the prices are discussed in the ‘Sensitivity Analysis’ section.

Capital Cost

There are several different ways of determining the capital costs of a plant. The three methods used in the determination of this process were: order of magnitude, calculation of individual pieces of equipment using the equation $\$ = A + bS^n$, and finally using the Aspen Economic Analyzer. Details about each method of determination will be explained in their respective section and compared at the end.

Order of Magnitude Method

There are several cost estimates that are made during a project with the roughest estimate being an order of magnitude estimate. This is performed to get a general idea of how much the plant will cost. The cost-curve method was not used due to lack of available information regarding a plant’s capital costs. The step count method was used, the cost is estimated by the Equation 1. We would like to produce 10,000 metric tons of MEK per year and the reactors have a 90% conversion. Because some units such as pumps and heat exchangers are not considered as functional units, our value of N is 8 because of the 3 reactors, 4 separation columns and the feed compressor. The cost was determined to be \$12.5 MM.

$$Cost = 380,000 N \left(\frac{Q}{s} \right)^{0.3} \quad (1)$$

Where C = ISBL capital cost in 2010 US dollars, Q = plant capacity in MT/yr., s = reactor conversion, N = number of functional units.

Hand Calculations

Another method used for determining the capital costs for the process are hand calculations using Equation 2. This approach uses estimates for the cost of individual unit operation equipment. This is made possible through knowing how the price of a piece of equipment changes when you change its characteristic dimension.

$$Cost (\$) = A + bS^n \quad (2)$$

where a, b = cost constants, S = characteristic dimension, and n = component specific exponent

A 'characteristic dimension' is the single most important piece of information about a specific process equipment when determining its purchase price. For example, distillation columns the characteristic dimension is the weight of the pressurized vessel and for a heat exchanger, it's the heat transfer area. This calculation was performed for all process equipment and the results are shown in Table 1. The total capital costs for all process equipment was determined to be \$6.924MM.

Table 1: Results for individual and total capital costs for the process using Hand Calculation method. Does not include the costs associated with the installation of equipment.

	Hand Calculations
Butanol Vaporizer	\$ 33,576.59
Reactor Feed Heater (2)	\$ 45,930.81
Reactors (3)	\$ 51,549.59
Condenser	\$ 45,930.81
Absorption Column	\$ 14,245.78
Extraction Column	\$ 17,229.40
Solvent Recovery Column	\$ 21,961.04
Recovery Column Reboiler	\$ 30,960.56
Recovery Column Condenser	\$ 5,650.00
Solvent Cooler	\$ 8,970.00
Product Purification Column	\$ 68,843.64
Product Column Reboiler	\$ 51,549.59
Product Column Condenser	\$ 42,641.17
Feed Compressor	\$ 1,641,858.76
Butanol Storage Tank	\$ 111,862.53
Solvent Storage Tank	\$ 14,641.11
Product Storage Tank	\$ 111,862.53
Total Equipment Cost	\$ 2,319,263.93

Aspen Calculations

The final method used for the determination of the purchase price of each MEK production unit is the Aspen Capital Cost Estimator. The full results which include the equipment and setting, piping, civil, instrumentation, paint and the labor cost can be found in Appendix I. In Figure 1, the initial setup for a typical piece of equipment is shown. Boxes outlined in red are required input for calculations.

Figure 1: Initial setup for AspenTech's Capital Cost Calculator.

After inputting specified information, the 'Evaluate' icon was clicked and calculations were performed inside the programs and details about the process equipment were displayed. The information includes construction material, dimensions of systems, design specification, and the associated costs of the equipment. Design data and costs shown in Tables(s) 2 and 3, respectively.

Table 2: Design Data for typical process equipment.

Parameter	Value	Units
Item type	JACKETED	
Material	SS	
Heat transfer area	2.0000	M2
Tube length	6.0003	M
Number of tubes per shell	1	
Design gauge pressure	1000.001	KPAG
Temperature	120.000	DEG C
Source of quote	SG	
Number of sections	1	
Total weight	440	KG

Table 3: Cost associated with the purchase of typical equipment. Not included are the costs associated with the installation of equipment.

Item	Material(USD)
Equipment&Setting	7800.
Piping	15426.
Civil	965.
Structural Steel	0.
Instrumentation	8785.
Electrical	0.
Insulation	3587.
Paint	382.
Subtotal	36945

Upon determining the purchasing price for each production unit, the costs were tabulated, and the results are shown in Table 4. The overall costs for each piece of equipment will be shown in the ISBL section. The installation costs for each piece of equipment varies from the method used in hand calculations.

Table 4: Results for individual and total capital costs for the process using AspenTech's Capital Cost Calculator. Does not include the costs associated with the installation of equipment.

	Aspen
Butanol Vaporizer	\$ 15,200.00
Reactor Feed Heater (2)	\$ 58,800.00
Reactors (3)	\$ 90,300.00
Condenser	\$ 26,400.00
Absorption Column	\$ 26,400.00
Extraction Column	\$ 24,600.00
Solvent Recovery Column	\$ 37,200.00
Recovery Column Reboiler	\$ 9,300.00
Recovery Column Condenser	\$ 2,800.00
Solvent Cooler	\$ 7,800.00
Product Purification Column	\$ 92,900.00
Product Column Reboiler	\$ 12,100.00
Product Column Condenser	\$ 13,400.00
Feed Compressor	\$ 1,223,600.00
Butanol Storage Tank	\$ 134,200.00
Solvent Storage Tank	\$ 24,000.00
Product Storage Tank	\$ 134,200.00
Total Equipment Cost	\$ 1,933,200.00

Cost Variations

There are noticeable differences in the prices of equipment for both the hand calculated prices versus those determined using AspenTech's software. There are a number of factors that could contribute to those differences. One of which is the difference in the level of accuracy, Aspen plus may be using a database that includes proprietary information not readily available to the public. They include costs of piping costs, civil costs, and instrumentation costs, these costs were not added to the cost of the equipment. Some of which would fall under the umbrella of installation costs which were purposely not included since the factors used for installation vary quite differently. One thing worth mentioning, neither Aspen or the hand calculations always price equipment consistently less or consistently higher when compared to the other. This observation is only valid for equipment that fall under a larger classification like heat exchangers, not specifics like double-pipe heat exchangers. Sample costs of some equipment are shown in Table 5. A full list of prices for comparison are shown in the Appendix.

Table 5: Comparisons between equipment prices. Does not include installation factors. Pricing for Hand calculations (2013) and Aspen calculations (2015). Small changes in CEPCI index indicates these values would be valid for both years.

	Hand Calculations	Aspen
Butanol Vaporizer	\$ 33,576.59	\$ 15,200.00
Reactor Feed Heater (2)	\$ 45,930.81	\$ 58,800.00
Feed Compressor	\$ 1,641,858.76	\$ 1,223,600.00
Butanol Storage Tank	\$ 111,862.53	\$ 134,200.00
Total Equipment Cost	\$ 2,319,263.93	\$ 1,933,200.00
2015 Costs Compared	\$ 2,344,528.24	\$ 1,933,200.00
Difference	\$	411,328.24

ISBL

The inside battery limits (ISBL) investment is the cost of the plant. The ISBL includes the equipment cost and installation. The cost of installation of each unit was accounted for in the ISBL. Figure 2 contains the installation factors for each unit.

Installation Factors Proposed by Hand (1958)	
Equipment Type	Installation Factor
Compressors	2.5
Distillation columns	4
Fired heaters	2
Heat exchangers	3.5
Instruments	4
Miscellaneous equipment	2.5
Pressure vessels	4
Pumps	4

Figure 2: List of installation factors for each process unit.

The total equipment and installed cost for each unit was determined to be \$6.92 MM and is outlined in Table 6. The installed costs were calculated using Equation 3.

$$C = F \sum C_e \quad (3)$$

Where C = total plant ISBL capital costs, F = installation factor or Lang factor, and C_e is the cost of individual equipment

	Hand Calculations
Butanol Vaporizer	\$ 117,518.07
Reactor Feed Heater (2)	\$ 321,515.69
Reactors (3)	\$ 386,621.96
Condenser	\$ 160,757.84
Absorption Column	\$ 51,681.67
Extraction Column	\$ 53,896.85
Solvent Recovery Column	\$ 60,354.69
Recovery Column Reboiler	\$ 123,842.26
Recovery Column Condenser	\$ 19,775.00
Solvent Cooler	\$ 31,395.00
Product Purification Column	\$ 208,684.55
Product Column Reboiler	\$ 180,423.58
Product Column Condenser	\$ 149,244.11
Feed Compressor	\$ 4,104,646.90
Butanol Storage Tank	\$ 447,450.11
Solvent Storage Tank	\$ 58,564.44
Product Storage Tank	\$ 447,450.11
Total Equipment Cost	\$ 6,923,822.85

OSBL

Per the problem statement, there are no off-site costs to be considered. Therefore, the outside battery limits (OSBL) cost is 0. For a typical plant, the OSBL costs include investments to infrastructure, such as steam mains, water pipes, laboratories, and power lines.

Contingency and Engineering Costs

The contingency costs are the additional costs that may arise from day to day plant operation. Contingency costs include changes in material prices, labor disputes, and other unexpected problems. The contingency cost was determined as a percentage of the total ISBL cost. The percentage was 10% as illustrated in Figure 3.

Typical Factors for Estimation of Project Fixed Capital Cost			
Item	Process Type		
	Fluids	Fluids-Solids	Solids
Major equipment, total purchase cost	C_e	C_e	C_e
f_{er} Equipment erection	0.3	0.5	0.6
f_p Piping	0.8	0.6	0.2
f_i Instrumentation and control	0.3	0.3	0.2
f_{el} Electrical	0.2	0.2	0.15
f_c Civil	0.3	0.3	0.2
f_s Structures and buildings	0.2	0.2	0.1
f_l Lagging and paint	0.1	0.1	0.05
ISBL cost, $C = \sum C_e \times$	3.3	3.2	2.5
Offsites (OS)	0.3	0.4	0.4
Design and Engineering (D&E)	0.3	0.25	0.2
Contingency (X)	0.1	0.1	0.1
Total fixed capital cost $C_{FC} = C (1 + OS)(1 + D\&E + X)$			
$= C \times$	1.82	1.89	1.82
$= \sum C_e \times$	6.00	6.05	4.55

Figure 3: Typical Factors for Estimation of Project Fixed Capital Costs

The design and engineering costs were also calculated as a percentage of the total ISBL cost, given in the above referenced table. The design and engineering costs include the cost to create and implement an engineering design, such as creating scale models, drafting, cost engineering, construction costs, and contractor fees. The design and engineering costs was based on 30% of the total ISBL.

The total fixed capital cost is the sum of the ISBL costs, OSBL costs, design and engineering costs, and contingency costs. The fixed capital costs are shown in Table 7. The total fixed capital cost was found to be \$9.69 MM.

The cost of working capital is also a factor of the total fixed capital cost. Working capital is the capital required to maintain daily plant operations. This includes capital for supplies, accounts receivable and payable, and spare parts. The working capital cost was based on 20% of the total fixed cost.

The total capital cost is the sum of the total fixed capital cost and the working capital and was found to be \$11.63 MM. The total capital cost the total investment cost of the plant.

Cost	\$MM/yr
Total ISBL Installed Cost	\$ 6.92
OSBL	\$ -
Design and Engineering	\$ 2.08
Contingency	\$ 0.69
Total Fixed Capital Cost	\$ 9.69
Working Capital	\$ 1.94

Table 7: Fixed Capital Costs

Variable Cost of Production

The variable cost of production is comprised of the raw material costs and the utility costs. The market cost of 2-butanol and trichloroethane is \$800/MT and \$1,150/MT respectively. Based on a production of 10,000 metric tons of MEK, the raw materials cost was calculated to be \$8.36 MM. The utility cost is comprised of electric, cooling water, steam, and fuel oil costs. The cost for electric, cooling water, low pressure steam, and fuel oil were found to be \$0.0708/kWh, \$0.0265/MT, \$9.81/MT, and \$7.60/GJ respectively.

The raw material and utility costs are listed in Table 8. The total variable cost of production was found to be \$10.08 MM.

Variable Cost of Production (\$MM / yr)	
Raw Materials	
2-Butanol	\$ 8.36
Trichloroethane	\$ 0.01
Total Raw Material Cost	\$ 8.37
Utilities	
Electrical	\$ 0.57
Cooling Water	\$ 0.03
LP Steam	\$ 0.09
Fuel Oil	\$ 1.03
Total Utility Cost	\$ 1.71
Variable Cost of Production	\$ 10.08

Table 8. Variable Cost of Production

Fixed Cost of Production

The fixed cost of production is comprised of the labor, maintenance, land/rent, insurance, overhead charges, taxes, and finance charges. The maintenance, land/rent, insurance, and tax costs were determined as a percentage of the ISBL costs: 3%, 1%, 1%, and 1% respectively. The labor cost was calculated based on 4.8 operators per shift position with 3 shift positions at \$73,000/year each. The cost for labor supervision is additional 25%. The direct overhead of labor and supervision is an additional 45% of the labor and supervision costs. The overhead charges are for administration personnel and was calculated at 65% of the labor, supervision, management, and maintenance costs. The fixed cost of production is listed in Table 9. The total fixed cost of production was determined to be \$3.77 MM.

Fixed Cost of Production (\$MM / yr)	
Labor Cost (Shift Position)	\$ 1.05
Supervisors	\$ 0.26
Overhead	\$ 0.59
Maintenance	\$ 0.29
Tax/Insurance	\$ 0.15
Overhead Charges	\$ 1.43
Total Fixed Cost of Production	\$ 3.77

Table 9: Fixed Cost of Production

Revenues and Gross Profit

Revenue consideration is an integral component of the feasibility to produce a product. The product revenue is the income received from the sale of the main product and any by-products. The market price of MEK was found to be \$2000/MT and the revenue of producing 10,000 metric tons of MEK was calculated to be \$20.00 MM. There are no sellable by-products of MEK production. The gross margin was calculated as the difference between the revenue and the total raw material cost. The gross margin shows range of profitability after accounting for the raw materials used to produce the product. The gross margin was found to be \$11.63 MM. The per unit gross margin is \$1163.20.

In addition to the gross margin, a variable contribution margin can be calculated. The variable contribution margin shows the process profitability on a fixed-cost-free basis. The variable contribution margin is calculated as the difference between the revenue and the variable cost of production: \$9.92 MM.

The cash cost of production is the cost of making a product. It is found by summing the total variable costs and the total fixed cost of production. The cash cost was found to be \$13.85 MM. From the cash cost of production, the gross profit can be calculated. The gross profit is the difference between the revenue and the cash cost of production. The gross profit was found to be \$6.15 MM. The gross profit is an estimate of the profitability after accounting for all of the costs of production.

To determine the profit after taxes, net profit is used. The net profit is the difference between the gross profit and the estimated taxes. The net profit was found to be \$6.01 MM. The revenue and profit estimations are listed in Table 10.

Overall	
Revenue (based on MEK \$2000/MT)	\$ 20.00
Gross Margin	\$ 11.63
Variable Contribution Margin	\$ 9.92
Cash Cost of Production	\$ 13.85
Gross Profit	\$ 6.15
Net Profit	\$ 6.01

Table 10. Revenue and Profit Estimations

Simple pay-back period and ROI

The simple payback time can be calculated to give an estimate of the time in years for the fixed capital costs to be paid back based on the gross profit. The simple payback time is the total fixed capital cost divided by the average annual cash flow. It assumes that the total capital cost is paid in year 0, as well as, revenues start generating immediately. It also a pretax estimate. The simple payback time was found to be 1.58 years. The post-tax simple payback time was found to be 1.61 years.

The return on investment can be calculated to determine the usefulness of a particular operation or process. It is the net annual profit divided by the total fixed capital cost. The return on investment was found to be 62%. Higher ROIs indicate a more lucrative investment as the potential profitability is high.

NPV

The net present value (NPV) for the production of MEK can be calculated to determine the present value of any future cash flows. The NPV is the difference between the present value of cash flows in and cash flows out of a project. It indicates whether a project will be profitable or experience loss over a given set of years. The NPV addresses the concept of the time value of money by presenting future profitability in terms of the value of today's money. The NPV is calculated via Equation 4.

$$NPV = \sum_{n=i}^{n=t} \frac{CF_n}{(1+i)^n} \quad (4)$$

Where CF_n = cash flow at year n, t is the project end year, i = interest rate.

The NPV was calculated with a discount rate of 14% and an estimated tax rate of 35%, using the MACRS 5 and 7- year recovery terms. The results are shown in Table 11 and Table 12.

MACRS is a depreciation method afforded by the US federal government. MACRS allows for the reallocation of plant depreciation according to different recovery periods. Plant depreciation provides tax benefits in terms of deductions, which lowers taxable income. Paying less taxes at the beginning of a plant operation increase a plant's cash flow. The MACRS 5-year term allocates the depreciation over a 5-year period. The MACRS 7-year term allocates the depreciation over a 7-year period.

As evidenced in the above referenced tables, the NPV at the end of 20 years using the MACRS the 5 and 7-year recovery terms is \$15.73 MM and \$15.57 MM respectively. This is a difference of about \$160,000. In general, the NPV for the 5-year term was greater than the NPV for the 7-year term. Based on the NPV results, the MACRS 5-year term would be more advantageous as it provides for greater profit than the 7-year term.

5 Year									
			Tax Rate	0.35	Interest	0.14			
Year	Pre-Tax CF	Depreciation	Taxable Income	Tax	CF	Fd	PV	NPV	MACRS
0	\$ (4.85)	\$ -	\$ (4.85)	\$ -	\$ (4.85)	1.00	\$ (4.85)	\$ (4.85)	0
1	\$ (4.85)	\$ -	\$ (4.85)	\$ -	\$ (4.85)	0.88	\$ (4.25)	\$ (9.10)	0
2	\$ 3.08	\$ 1.94	\$ 1.14	\$ -	\$ 3.08	0.77	\$ 2.37	\$ (6.73)	20
3	\$ 6.15	\$ 3.10	\$ 3.05	\$ 0.40	\$ 5.75	0.67	\$ 3.88	\$ (2.85)	32
4	\$ 6.15	\$ 1.86	\$ 4.29	\$ 1.07	\$ 5.08	0.59	\$ 3.01	\$ 0.16	19.2
5	\$ 6.15	\$ 1.12	\$ 5.03	\$ 1.50	\$ 4.65	0.52	\$ 2.41	\$ 2.58	11.52
6	\$ 6.15	\$ 1.12	\$ 5.03	\$ 1.76	\$ 4.39	0.46	\$ 2.00	\$ 4.58	11.52
7	\$ 6.15	\$ 0.56	\$ 5.59	\$ 1.76	\$ 4.39	0.40	\$ 1.75	\$ 6.33	5.76
8	\$ 6.15	\$ -	\$ 6.15	\$ 1.96	\$ 4.19	0.35	\$ 1.47	\$ 7.80	0
9	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.31	\$ 1.23	\$ 9.03	0
10	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.27	\$ 1.08	\$ 10.11	0
11	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.24	\$ 0.95	\$ 11.05	0
12	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.21	\$ 0.83	\$ 11.88	0
13	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.18	\$ 0.73	\$ 12.61	0
14	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.16	\$ 0.64	\$ 13.25	0
15	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.14	\$ 0.56	\$ 13.81	0
16	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.12	\$ 0.49	\$ 14.30	0
17	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.11	\$ 0.43	\$ 14.73	0
18	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.09	\$ 0.38	\$ 15.11	0
19	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.08	\$ 0.33	\$ 15.44	0
20	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.07	\$ 0.29	\$ 15.73	0

Table 11: NPV with MACRS 5-Year

7 Year									
Year	Pre-Tax CF	Depreciation	Tax Rate	0.35		Interest		0.14	
			Taxable Income	Tax	CF	Fd	PV	NPV	MACRS
0	\$ (4.85)	\$ -	\$ (4.85)	\$ -	\$ (4.85)	1.00	\$ (4.85)	\$ (4.85)	0
1	\$ (4.85)	\$ -	\$ (4.85)	\$ -	\$ (4.85)	0.88	\$ (4.25)	\$ (9.10)	0
2	\$ 3.08	\$ 1.39	\$ 1.69	\$ -	\$ 3.08	0.77	\$ 2.37	\$ (6.73)	14.29
3	\$ 6.15	\$ 2.37	\$ 3.78	\$ 0.59	\$ 5.56	0.67	\$ 3.75	\$ (2.98)	24.49
4	\$ 6.15	\$ 1.70	\$ 4.46	\$ 1.32	\$ 4.83	0.59	\$ 2.86	\$ (0.12)	17.49
5	\$ 6.15	\$ 1.21	\$ 4.94	\$ 1.56	\$ 4.59	0.52	\$ 2.38	\$ 2.26	12.49
6	\$ 6.15	\$ 0.87	\$ 5.29	\$ 1.73	\$ 4.42	0.46	\$ 2.01	\$ 4.28	8.93
7	\$ 6.15	\$ 0.86	\$ 5.29	\$ 1.85	\$ 4.30	0.40	\$ 1.72	\$ 6.00	8.92
8	\$ 6.15	\$ 0.87	\$ 5.29	\$ 1.85	\$ 4.30	0.35	\$ 1.51	\$ 7.51	8.93
9	\$ 6.15	\$ 0.43	\$ 5.72	\$ 1.85	\$ 4.30	0.31	\$ 1.32	\$ 8.83	4.46
10	\$ 6.15	\$ -	\$ 6.15	\$ 2.00	\$ 4.15	0.27	\$ 1.12	\$ 9.95	0
11	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.24	\$ 0.95	\$ 10.89	0
12	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.21	\$ 0.83	\$ 11.72	0
13	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.18	\$ 0.73	\$ 12.45	0
14	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.16	\$ 0.64	\$ 13.09	0
15	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.14	\$ 0.56	\$ 13.65	0
16	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.12	\$ 0.49	\$ 14.14	0
17	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.11	\$ 0.43	\$ 14.57	0
18	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.09	\$ 0.38	\$ 14.95	0
19	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.08	\$ 0.33	\$ 15.28	0
20	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.07	\$ 0.29	\$ 15.57	0

Table 12: NPV with MACRS 7-Year

DCFROR with MACRS Depreciation

The discounted cash flow rate of return is a tool used when determining which process is more profitable. This is achieved through an equation similar to Equation 4, with a slight change in your results. Instead of getting the NPV value after a variable amount of years, you change the interest rate for the process so that the NPV is \$0 at the end of the process cycle, in our example, this is twenty years. The equation is shown in Equation 5.

$$\sum_{n=i}^{n=t} \frac{CF_n}{(1+i)^n} = 0 \quad (5)$$

The process of finding the correct interest rate is achieved through iteration, which is made possible through Excel's Goal Seek function. Results from the analysis performed on the 5-year MACRS cycle is shown in Table 13.

5 Year									
Year	Pre-Tax CF	Depreciation	Tax Rate	0.35		Interest		0.37897439	
			Taxable Income	Tax	CF	Fd	PV	NPV	MACRS
0	\$ (4.85)	\$ -	\$ (4.85)	\$ -	\$ (4.85)	1.00	\$ (4.85)	\$ (4.85)	0
1	\$ (4.85)	\$ -	\$ (4.85)	\$ -	\$ (4.85)	0.73	\$ (3.51)	\$ (8.36)	0
2	\$ 3.08	\$ 1.94	\$ 1.14	\$ -	\$ 3.08	0.53	\$ 1.62	\$ (6.74)	20
3	\$ 6.15	\$ 3.10	\$ 3.05	\$ 0.40	\$ 5.75	0.38	\$ 2.19	\$ (4.55)	32
4	\$ 6.15	\$ 1.86	\$ 4.29	\$ 1.07	\$ 5.08	0.28	\$ 1.41	\$ (3.14)	19.2
5	\$ 6.15	\$ 1.12	\$ 5.03	\$ 1.50	\$ 4.65	0.20	\$ 0.93	\$ (2.21)	11.52
6	\$ 6.15	\$ 1.12	\$ 5.03	\$ 1.76	\$ 4.39	0.15	\$ 0.64	\$ (1.57)	11.52
7	\$ 6.15	\$ 0.56	\$ 5.59	\$ 1.76	\$ 4.39	0.11	\$ 0.46	\$ (1.11)	5.76
8	\$ 6.15	\$ -	\$ 6.15	\$ 1.96	\$ 4.19	0.08	\$ 0.32	\$ (0.79)	0
9	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.06	\$ 0.22	\$ (0.57)	0
10	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.04	\$ 0.16	\$ (0.41)	0
11	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.03	\$ 0.12	\$ (0.29)	0
12	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.02	\$ 0.08	\$ (0.21)	0
13	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.02	\$ 0.06	\$ (0.14)	0
14	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.01	\$ 0.04	\$ (0.10)	0
15	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.01	\$ 0.03	\$ (0.07)	0
16	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.01	\$ 0.02	\$ (0.04)	0
17	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.00	\$ 0.02	\$ (0.03)	0
18	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.00	\$ 0.01	\$ (0.02)	0
19	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.00	\$ 0.01	\$ (0.01)	0
20	\$ 6.15	\$ -	\$ 6.15	\$ 2.15	\$ 4.00	0.00	\$ 0.01	\$ 0.00	0

Table 13: Results from the DCFOR with a 5-year depreciation cycle.

The interest rate for the MACRS 5-year cycle is 37.9% while the interest rate for the MACRS 7-year cycle is 37.4%. The results prove that the 5-year cycle is more profitable and should be the approach taken when structuring depreciation.

Sensitivity Analysis

Although the previous sections estimate the cost of the plant based on the material and equipment, there are several other factors such as the cost of utilities, raw materials and product which must also be taken into consideration when deciding if the plant should be built. The sensitivity analysis determines the risk that is associated with undertaking such a project. Since the market for the product and raw materials may fluctuate, these changes were mapped and shown in Figure 4.

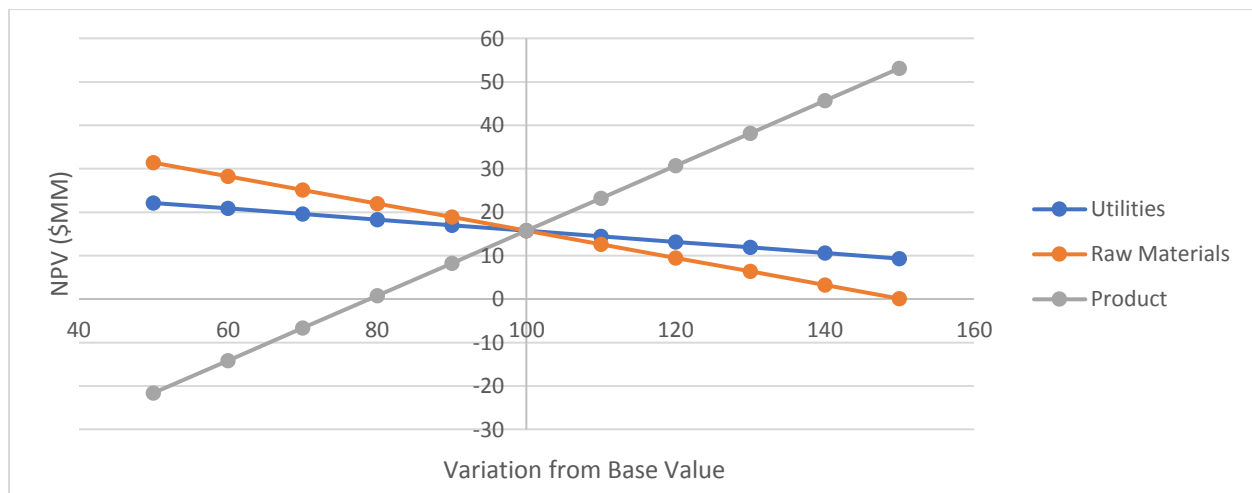


Figure 4: Changes in Market graphed on basis change.

Appendix I

Table 13. Dimensions for Process Equipment developed through AspenTech's Capital Cost Calculator

Parameter	Value	Units
Item type	FIXED T S	
Number of identical items	3	
GENERAL DESIGN DATA		
TEMA type	BEM	
Heat exchanger design option	STAND	
Heat exchanger design+cost tool	ECON	
Heat transfer area	50.000	M2
Number of shells	1	
Number of tube passes	1	
Number of shell passes	1	
Vendor grade	HIGH	
SHELL DATA		
Shell material	A285C	
Shell diameter	400.002	MM
Shell length	5.0000	M
Shell design gauge pressure	500.000	KPAG
Shell design temperature	340.000	DEG C
Shell operating temperature	340.000	DEG C
Shell corrosion allowance	0.0	MM
Shell wall thickness	9.0002	MM
ASA rating Shell side	150	CLASS
Number of baffles	14	
Shell fabrication type	PIPE	
Expansion joint	NO	
TUBE DATA		
Tube material	A 214	
Number of tubes per shell	126	

Table 14: Example of complete cost summary for process equipment.

Item	Material(USD)	Manpower(USD)	Manhours
Equipment&Setting	90300.	3258.	106
Piping	81741.	39546.	1261
Civil	3044.	4640.	185
Structural Steel	0.	0.	0
Instrumentation	32660.	12073.	381
Electrical	0.	0.	0
Insulation	25533.	20439.	860
Paint	810.	2114.	90
Subtotal	234088	82070	2883

Total material and manpower cost=USD 316200.

Table 14: All Process Equipment price determination using Hand Calculation Method.

	Absorption Column		Extraction Column		Solvent Recovery Column		Product Purification Column		Solvent Storage Tank
Type	Vertical	Ceramic Saddles	Vertical	SS Pall Rings	Vertical	Sieve Plates	Vertical	Sieve Plates	Horizontal
Inside Diameter (m)	0.50	0.50	0.50	0.50	0.60	0.60	1.00	1.00	1.50
Height (m)	6.00	4.50	4.00	3.00	6.00		20.00		5.00
Volume (m³)		0.88		0.59					
Pressure (bar)	2.00		2.00		2.00		2.00		2.00
Pressure with 10% (bar)	2.20		2.20		2.20		2.20		2.20
S (ksi)	12.90		12.90		12.90		12.90		12.90
Thickness (m)	6.19E-04		6.19E-04		7.43E-04		1.24E-03		1.86E-03
Outside Diameter (m)	0.50		0.50		0.60		1.00		1.50
Weight (kg)	45.87		30.58		66.06		611.65		344.05
Heating Surface Area (m²)									
Power (kW)									
A	11600		11600		11600	130	17400	130	10200
b	34	2000	34	8500	34	440	79	440	31
S	45.87	0.88	30.58	0.59	66.06	0.6	611.65	1	344.05
n	0.85	1	0.85	1	0.85	1.8	0.85	1.2	0.85
Stainless Steel Factor	1	1	1	1	1	1	1.3	1.3	1
Equipment Cost	\$ 12,478.63	\$ 1,767.15	\$ 12,222.49	\$ 5,006.91	\$ 12,797.88	\$ 9,163.16	\$ 46,613.64	\$ 22,230.00	\$ 14,641.11
Installation Factor	4	1	4	1	4	1	4	1	4
# of Equipment	1	1	1	1	1	30	1	30	1
Total Cost	\$ 51,681.67	\$ 51,681.67	\$ 53,896.85	\$ 53,896.85	\$ 60,354.69	\$ 60,354.69	\$ 208,684.55	\$ 208,684.55	\$ 58,564.44

	Butanol Vaporizer	Reactor Feed Heater	Reactor	Condenser	Recovery Column Reboiler	Recovery Column Condenser
Type	S&T Kettle	S&T Fixed Head	S&T Fixed Tube	S&T Fixed Sheet	Thermosiphon	Double Pipe
Volume (m³)						
Heating Surface Area (m²)	15.0	25.0	50.0	25.0	4.0	1.5
Power (kW)						
A	29000	32000	32000	32000	30400	1900
b	400	70	70	70	122	2500
S	15	25	50.0	25	4	1.5
n	0.9	1.2	1.2	1.2	1.1	1
Stainless Steel Factor	1	1.3	1.3	1.3	1	1
Equipment Cost	\$ 33,576.59	\$ 45,930.81	\$ 51,549.59	\$ 45,930.81	\$ 30,960.56	\$ 5,650.00
Installation Factor	3.5	3.5	2.5	3.5	4	3.5
# of Equipment	1	2	3	1	1	1
Total Cost	\$ 117,518.07	\$ 321,515.69	\$ 386,621.96	\$ 160,757.84	\$ 123,842.26	\$ 19,775.00

	Solvent Cooler	Product Column Reboiler	Product Column Condenser	Feed Compressor	Butanol Storage Tank	Product Storage Tank
Type	Double Pipe	Kettle	S&T Floating	Centrifugal	Cone Roof	Cone Roof
Volume (m³)						
Heating Surface Area (m²)	2.0	4.0	15		400	400
Power (kW)				750		
A	1900	32000	32000	580000	5800	5800
b	2500	70	70	20000	1600	1600
S	2	50	15	750.00	400.00	400.00
n	1	1.2	0.9	0.6	0.7	0.7
Stainless Steel Factor	1.3	1.3	1.3	1	1	1
Equipment Cost	\$ 8,970.00	\$ 51,549.59	\$ 42,641.17	\$ 1,641,858.76	\$ 111,862.53	\$ 111,862.53
Installation Factor	3.5	3.5	3.5	2.5	4	4
# of Equipment	1	1	1	1	1	1
Total Cost	\$ 31,395.00	\$ 180,423.58	\$ 149,244.11	\$ 4,104,646.90	\$ 447,450.11	\$ 447,450.11

Table 15: Comparisons between Hand Calculations and Aspen Calculations for all Equipment.

	Hand Calculations	Aspen
Butanol Vaporizer	\$ 117,518.07	\$ 66,900.00
Reactor Feed Heater (2)	\$ 321,515.69	\$ 326,200.00
Reactors (3)	\$ 386,621.96	\$ 948,600.00
Condenser	\$ 160,757.84	\$ 83,200.00
Absorption Column	\$ 51,681.67	\$ 117,400.00
Extraction Column	\$ 53,896.85	\$ 125,700.00
Solvent Recovery Column	\$ 60,354.69	\$ 149,300.00
Recovery Column Reboiler	\$ 123,842.26	\$ 44,100.00
Recovery Column Condenser	\$ 19,775.00	\$ 52,000.00
Solvent Cooler	\$ 31,395.00	\$ 57,000.00
Product Purification Column	\$ 208,684.55	\$ 260,800.00
Product Column Reboiler	\$ 180,423.58	\$ 61,000.00
Product Column Condenser	\$ 149,244.11	\$ 80,500.00
Feed Compressor	\$ 4,104,646.90	\$ 1,545,300.00
Butanol Storage Tank	\$ 447,450.11	\$ 225,800.00
Solvent Storage Tank	\$ 58,564.44	\$ 141,100.00
Product Storage Tank	\$ 447,450.11	\$ 225,800.00
Total Equipment Cost	\$ 6,923,822.85	\$ 4,073,100.00
Year of Calculation	2010	2015
M&S Index	550.8	556.8
2019	603.1	603.1
2019 Equipment Costs	\$ 7,581,259.19	\$ 4,411,793.48

Sample conversion calculations:

The cost is in 2010 dollars, which will be converted into 2018 dollars using Equation 2 and the Chemical Engineering Plant Cost Index (CEPCI) for both years.

$$Cost\ in\ 2018 = Cost\ in\ 2010 \times \frac{CEPCI\ 2018}{CEPCI\ 2010} \quad (2)$$

Where CEPCI 2010 = 532.9, CEPCI 2018 = 603.1